

Group Recommender Systems

Exercise 5, May 11, 2021

1 Overview

During this exercise you will apply preferences aggregation strategies to recommend items to a group of users.

Disclaimer: This exercise is based on the exercise designed originally by Prof. Lorena Recalde of the UTM University, Ecuador.

2 Learning Objectives

- Understand methods for group recommendations.
- Understand the differences between aggregation strategies.

3 Instructions

In Moodle, you will find an Excel file called `GRS_dataset.xlsx` which contains the following sheets:

- **Movies Vector representation:** Attributes of the movies which will be recommended to the groups and the attributes of the movies already rated by the group members.
- **Similarity RatedMovies & NewMovn:** Cosine similarity between the movies already rated and the movies to be recommended.
- **Users-Movie Ratings:** Ratings given by the users to the movies (up) and the same matrix without empty values (down). The empty positions were filled using the average of rating strategy.
- **Example:** An example of an aggregation strategy used to get recommendations to a given group.

Proceed to use the data from the file to solve the following tasks, you are free to use Excel or Python to solve it.

Table 1:

Groups	
Group ID	User IDs in the group
1	10, 26, 30, 12, 11, 16, 37, 29, 36
2	2, 3, 28, 44, 41
3	11, 15, 29
4	22, 50, 48
5	31, 32, 33, 34, 40, 42, 18, 21, 48
6	7, 8, 24, 25, 44
7	1, 6, 29
8	14, 27, 48
9	23, 10, 16, 36, 48
10	4, 20, 28, 47 46, 45, 39, 49, 19
11	17, 2, 43, 5, 47
12	20, 28, 48, 35, 4
13	1, 51, 52, 53, 54, 55, 56, 57, 58
14	4, 19, 45, 46, 49
15	39, 45, 47

Besides a document with the answers, you must upload in Moodle the Excel file or Python script used to find the solutions.

4 Task: Recommending movies to groups

From the groups displayed on Table 1, choose two.

For the two selected groups, select and apply three aggregation strategies, get recommendations and present them as a ranking (Top 3) per each group. Table 2 shows an example of the format of the expected answer.

Table 2:

Group ID	Strategy	Recommendation		
		1st.	2nd.	3rd.
	X			
	Y			
	Z			

Answer the following questions:

- Are there any differences between the recommendation lists that you obtained? Why are those changes? which one do you think is better?
- The groups participated in a real experiment. The decisions they made are presented in Table 3, you can find the names of the movies corresponding to each ID in the sheet **Movies Vector Representation** (Movies to Recommend).

Are these results different from the ones you obtained? What do you think are the reasons for that? Which advantages and disadvantages can you identify from the different aggregation strategies?

Table 3:

Group ID	Recommendation		
	1st.	2nd.	3rd.
1	3	2	13
2	11	3	4
3	2	3	11
4	6	3	15
5	15	3	9
6	11	3	2
7	13	5	8
8	6	15	2
9	9	3	2
10	5	3	13
11	5	1	2
12	3	5	6
13	1	8	11
14	6	5	14
15	13	5	6